

Cybersecurity Training Diary

Day 4: OSI & TCP/IP Models

1. Topics Covered

✓ OSI Model (7 Layers)

Layer	Name	Function	Example Protocols	Security Concerns
7	Application	User interface	HTTP, FTP, SMTP	Phishing, API attacks
6	Presentation	Data translation/encryption	SSL, TLS	Weak encryption
5	Session	Manages connections	NetBIOS	Session hijacking
4	Transport	End-to-end communication	TCP, UDP	SYN floods, Port scanning
3	Network	Logical addressing/routing	IP, ICMP	IP spoofing
2	Data Link	Physical addressing	MAC, Ethernet	MAC spoofing
1	Physical	Raw bit transmission	Cables, WiFi	Cable tapping

✓ TCP/IP Model (4 Layers)

Layer	OSI Equivalent	Protocols	Key Function
Application	App, Presentation, Session	HTTP, DNS, FTP	User data
Transport	Transport	TCP, UDP	Reliable delivery
Internet	Network	IP, ICMP	Routing
Network Access	Data Link, Physical	Ethernet, WiFi	Physical transmission

✓ Key Differences

- **OSI:** Theoretical (7 layers), helps troubleshoot networks
- **TCP/IP:** Practical (4 layers), what the Internet actually uses

2. Key Takeaways

OSI is for learning, TCP/IP is for doing

Attacks happen at every layer (e.g., Physical = cable cutting, Application = SQL injection)

Encryption spans multiple layers (SSL at Presentation, IPsec at Network)

3. Practical Exercises

Task 1: Map Real-World Traffic to OSI Layers

1. Visit a website in your browser
2. Identify which OSI layers are involved:
 - Physical (WiFi/Ethernet)
 - Network (IP)
 - Transport (TCP)
 - Application (HTTP)

Task 2: Use `tcpdump` to View Layers

```
tcpdump -i any -n -v # View raw network traffic by layer
```

4. Questions for Reflection

Why doesn't the Internet use the OSI model?

At which layer would you implement a firewall? (Hint: Multiple answers possible)

Name - Bhawanpreet Kaur

URN - 2302499